

Monohybrid Cross & Mendel's Laws

Worksheet · PU-II Biology, Chapter 4

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Worksheet – Monohybrid Cross & Mendel's Laws

15 questions · answer key on the last page

Q1. In a monohybrid cross TT (tall) x tt (dwarf), the F₁ generation is:

- (a) All tall (Tt)
- (b) All dwarf (tt)
- (c) 3 tall : 1 dwarf
- (d) 1 tall : 1 dwarf

Q2. On selfing the F₁ (Tt x Tt), the F₂ phenotypic ratio for tall:dwarf is:

- (a) 1:1
- (b) 1:2:1
- (c) 3:1
- (d) 9:3:3:1

Q3. The F₂ genotypic ratio in a monohybrid cross (TT:Tt:tt) is:

- (a) 1:1:1
- (b) 1:2:1
- (c) 3:1
- (d) 9:3:3:1

Q4. Who devised the graphical grid method for predicting the genotypes of a cross?

- (a) Gregor Mendel
- (b) Reginald C. Punnett
- (c) Thomas Hunt Morgan
- (d) Walter Sutton

Q5. Mendel's Law of Dominance is based on which type of cross?

- (a) Dihybrid cross
- (b) Monohybrid cross
- (c) Test cross
- (d) Trihybrid cross

Q6. Which law states that the two alleles of a gene pair separate during gamete formation so that each gamete receives only one allele?

- (a) Law of Dominance
- (b) Law of Segregation
- (c) Law of Independent Assortment
- (d) Law of Recombination

Q7. The Law of Segregation is also known as the:

- (a) Law of Purity of Gametes
- (b) Law of Dominance
- (c) Law of Independent Assortment
- (d) Law of Linkage

Q8. In coat colour inheritance in guinea pigs, which allele is dominant?

- (a) White (b)
- (b) Black (B)
- (c) Neither is dominant
- (d) Both are co-dominant

Q9. A homozygous parent (e.g. TT) produces gametes that are:

- (a) 50% T, 50% t

- (b) All of one type only (T)
- (c) All heterozygous
- (d) 25% T, 75% t

Q10. A heterozygous parent (Tt) produces gametes in what proportion?

- (a) All T
- (b) All t
- (c) Two types, T and t, in equal proportion
- (d) Three types in unequal proportion

Q11. Mendel's First Law explains why:

- (a) Both parental characters appear equally in F1
- (b) Only one parental character is seen in F1, while both appear in F2
- (c) F1 always shows an intermediate character
- (d) F2 shows only the recessive character

Q12. In the Punnett square method, gametes are conventionally arranged:

- (a) Only along the top row
- (b) Only along the left column
- (c) Along the top row and left column
- (d) Diagonally across the square

Q13. The F2 ratio $TT:Tt:tt = 1:2:1$ can be derived from the binomial expansion of which expression, where a and b are gamete frequencies of half each, and x, y are the dominant and recessive alleles?

- (a) $(ax+by)$
- (b) $(ax+by)$ squared
- (c) $(ax-by)$ squared
- (d) $(a+b)x$

Q14. In a test cross, an unknown tall pea plant is crossed with a homozygous recessive dwarf plant (tt), and ALL the progeny are tall. The unknown parent's genotype must be:

- (a) tt
- (b) Tt
- (c) TT
- (d) Cannot be determined

Q15. In a test cross, an unknown tall pea plant is crossed with tt, and the progeny obtained is 1 tall : 1 dwarf. The genotype of the unknown parent is:

- (a) TT
- (b) Tt
- (c) tt
- (d) Cannot be determined without a further cross

Answer Key & Explanations

- Q1.** (a) All tall (Tt) — Crossing homozygous tall (TT) with homozygous dwarf (tt) gives an F₁ that is entirely tall and heterozygous (Tt), since tall is dominant.
- Q2.** (c) 3:1 — Selfing heterozygous F₁ (Tt x Tt) gives an F₂ phenotypic ratio of 3 tall : 1 dwarf, as the recessive dwarf trait reappears.
- Q3.** (b) 1:2:1 — The F₂ genotypic ratio from a Tt x Tt cross is 1 TT : 2 Tt : 1 tt, even though the phenotypic ratio is 3:1.
- Q4.** (b) Reginald C. Punnett — The British geneticist Reginald C. Punnett devised the Punnett square, a grid method for calculating the probability of genotypes in a cross.
- Q5.** (b) Monohybrid cross — The Law of Dominance, along with the Law of Segregation, is based on Mendel's monohybrid cross experiments.
- Q6.** (b) Law of Segregation — The Law of Segregation (Law of Purity of Gametes) states that the two alleles of a gene pair separate during gamete formation.
- Q7.** (a) Law of Purity of Gametes — The Law of Segregation is also called the Law of Purity of Gametes because each gamete carries only one, 'pure' allele of a gene pair.
- Q8.** (b) Black (B) — In guinea pigs (*Cavia porcellus*), the black coat colour allele (B) is dominant over the white allele (b).
- Q9.** (b) All of one type only (T) — A homozygous parent like TT can only produce gametes carrying the T allele, since both its alleles are identical.
- Q10.** (c) Two types, T and t, in equal proportion — A heterozygous Tt parent produces two types of gametes, T and t, in equal proportion, as per the Law of Segregation.
- Q11.** (b) Only one parental character is seen in F₁, while both appear in F₂ — The Law of Dominance explains why F₁ resembles only one parent (the dominant one), while both the dominant and recessive traits reappear in F₂.
- Q12.** (c) Along the top row and left column — In a Punnett square, the gametes of one parent are listed along the top row and those of the other parent along the left column.
- Q13.** (b) (ax+by) squared — Expanding (ax+by) squared gives a-squared x-squared + 2ab xy + b-squared y-squared, matching the 1:2:1 genotypic ratio (TT:Tt:tt).
- Q14.** (c) TT — If all progeny of the test cross are tall, the unknown parent must be homozygous dominant (TT), since it can only pass on the T allele.
- Q15.** (b) Tt — A 1:1 tall to dwarf ratio in the test cross progeny shows the unknown parent is heterozygous (Tt), producing both T and t gametes.